

Concurrency and Distributed Systems

Week 9 – Part 4: Knight Capital and Phase 3 Bridge

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Part 4 Roadmap

1 War Story

2 Phase 3 Bridge

Knight Capital Discussion

- Timing and deployment can create system-level failure.
- Stale code paths can be reactivated accidentally.
- Operational controls limit blast radius.
- Concurrency thinking extends beyond locks and threads.

Discussion Questions

- ① What was the local technical failure?
- ② What was the operational failure?
- ③ What control could have reduced the blast radius?
- ④ How is this similar to race-type thinking?

From Runtime Limits to Distributed Systems

Week 10 shift

Networked systems remove shared memory and add partial failure.

- A network is a slow queue with failure.
- Retries can amplify load.
- Measurement remains mandatory.

Run

```
.\run-demo.ps1 04
```

Use the case-study handout: `case-studies/W9_Knight-Capital-2012.md`.